Graphical Perception Across Encoding Channels in Multimodal LLMs

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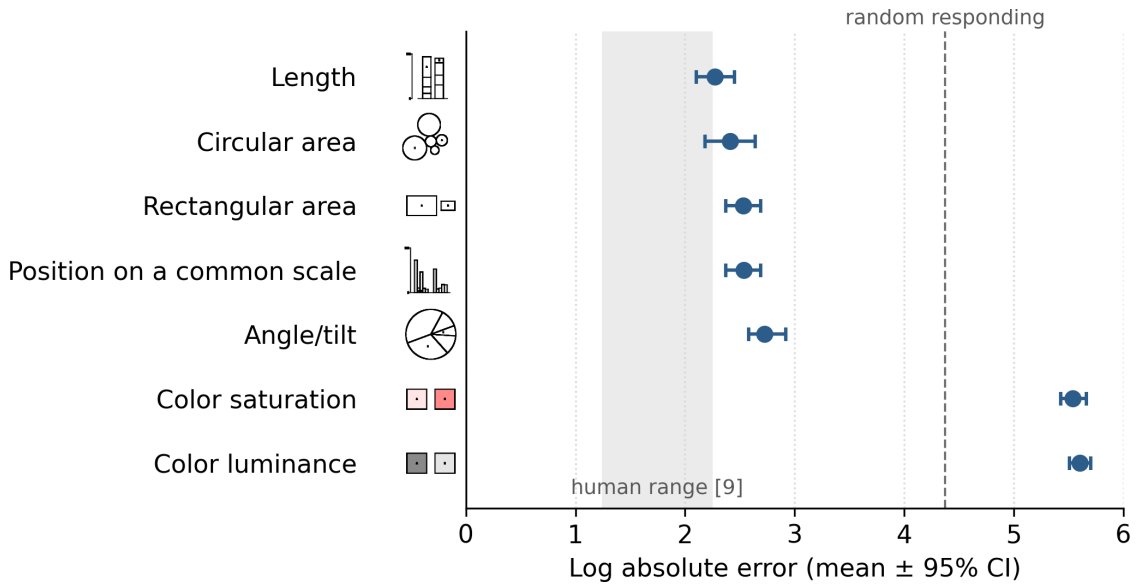


Figure 1: Mean log absolute error (LAE) per encoding channel, aggregated over six MLLMs and six prompting conditions; lower is better, and a one-unit decrease halves the estimation error. The shaded band spans the per-task human means reported by Heer and Bostock [9], and the dashed line marks random responding (4.37). The five geometric channels span 0.45 LAE units with no significant pairwise difference, while both color channels exceed random responding.

Abstract

Visual encoding channels are ranked for human observers by how accurately people decode quantities from them, and that ranking underpins decades of design guidance. We ask whether it carries over to multimodal LLMs. We replicate the Cleveland–McGill and Heer–Bostock graphical perception experiments across six models from three capability tiers under six prompting conditions, and extend the task set with two color-based channels. We report three findings. The geometric channels are interchangeable, with no significant pairwise difference, consistently across models and prompts. Luminance and saturation are not usable quantitative channels at all: errors exceed those expected from random responding. A follow-up experiment shows that distractor elements degrade estimation, whereas spatial separation between the compared marks does not, suggesting that for agent-facing charts, the amount of surrounding visual material matters more than the choice of encoding.

Index Terms: Graphical perception, multimodal large language models, visual encoding channels

1 Introduction

The effectiveness of a visualization rests on how data is mapped to visual properties. Cleveland and McGill [4] put that intuition on an empirical footing by measuring how accurately people extract quantities from different encodings, producing a ranking that places position on a common scale above length, angle, area, and finally color. Heer and Bostock [9] replicated and extended the result at a crowdsourced scale, and the ranking has since been consolidated into design guidance [12] and encoded directly into automated chart design systems [11].

Charts are no longer read only by people. Multimodal large language models (MLLMs) are embedded in tools that summarize dashboards, answer questions about figures and act as agents in analysis pipelines with no human in the loop. If the accuracy differences that motivate our design guidance do not hold for these systems, then charts optimized for human perception are being tuned on the wrong objective for a growing share of their readers.

Two recent studies ask this question directly and reach conclusions that are hard to reconcile. Guo et al. [6] evaluate GPT-4o-mini on seven of the original tasks and find that its judgments show no statistical difference across tasks, unlike the systematic differences humans exhibit; their limitations section names single-model evaluation as the main threat to generality. Nguyen et al. [13] evaluate four pretrained and several fine-tuned MLLMs across five Cleveland–McGill experiments and report that models outperform the human baseline. As in these earlier studies, we replicate the Cleveland–McGill experiment with six MLLMs and report three main findings.

- *Geometric encodings are interchangeable for MLLMs.* Position, length, angle, circular area and rectangular area produce statistically indistinguishable errors, generalizing the single-model observation of Guo et al. [6] to six architecturally diverse models and to all six of our prompting conditions.
- *Color is not a quantitative channel.* Luminance and saturation, evaluated here for the first time in this task family, yield errors worse than those of random responding.

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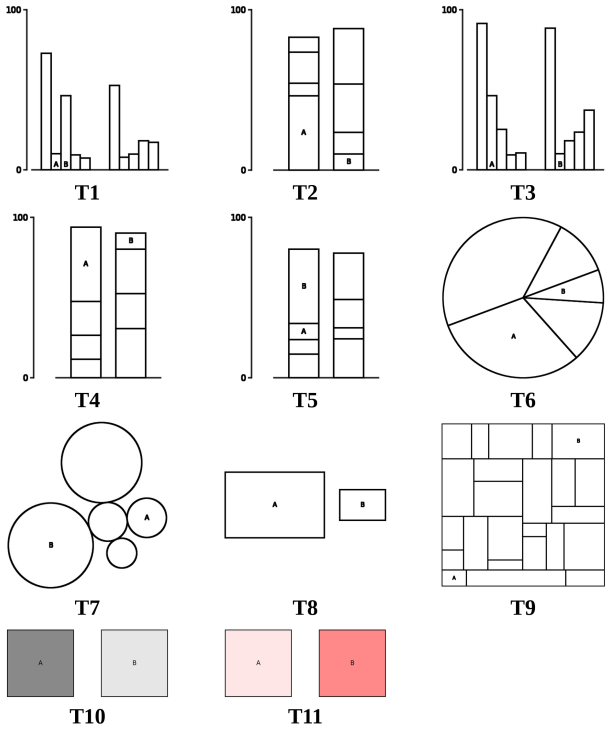


Figure 2: Representative stimuli for the eleven tasks. The two elements to be compared are labelled *A* and *B* in every task.

- *Surrounding visual elements matter.* Moving the compared marks apart leaves estimation error unchanged, so the elevated errors on our two most cluttered tasks are attributable to the intervening distractor elements rather than to distance.

2 Related Work

Human graphical perception. Bertin [3] laid out visual variables as a theoretical framework, and Cleveland and McGill [4] made the ranking empirical through elementary perceptual tasks in which participants judge the proportion between two marked elements. Ware [14] grounded the ordering in pre-attentive processing, and Munzner [12] consolidated it into channel rankings by data type. Heer and Bostock [9] replicated the original experiments on Mechanical Turk and added rectangular area judgments of the kind used in treemaps [1]. Their published per-task error profile is the human baseline we compare against throughout.

Machine graphical perception. Haehn et al. [7] first ran these stimuli through CNNs and found that networks that excel on general vision benchmarks do not reproduce the human ranking. Cui et al. [5] extended this to relational reasoning on bar charts. This line of work establishes that the human ranking is not automatically inherited by a visual system trained on natural images.

Bendeck and Stasko [2] evaluate GPT-4 on visualization literacy tests and report weakness on value retrieval and color distinctions. Han et al. [8] benchmark low-level chart question answering and find frontier accuracy well short of the ceiling. Guo et al. [6] and Nguyen et al. [13] replicate the graphical perception experiments directly, as described above. We differ from both in three respects: we add color-based encoding channels, we hold stimuli and prompts fixed across an architecturally diverse set of models, and we isolate spatial separation as an independent variable rather than confounding it with the surrounding chart structure.

Table 1: The eleven tasks and the seven channels they cover. Tasks 1–9 follow Heer and Bostock [9]; Tasks 10–11 are new.

Task		Channel
T1	bar chart, adjacent bars	position on common scale
T2	stacked bars, bottom segments	position on common scale
T3	bar chart, separate groups	position on common scale
T4	stacked bars, top segments	length
T5	stacked bars, one bar	length
T6	pie chart	angle/tilt
T7	bubble chart	circular area
T8	two isolated rectangles	rectangular area
T9	treemap	rectangular area
T10	two squares	color luminance
T11	two squares	color saturation

3 Method

We replicate the Cleveland–McGill [4] graphical perception experiment with MLLMs. Each stimulus marks two elements, A and B, and the model is asked what percentage the smaller is of the larger. We use eleven task types, each pairing a visual encoding channel with a chart type. Per task type, we generate 45 stimuli whose true ratios span nine levels between 0.18 and 0.83, and evaluate six models under six prompt variants, scoring responses by log absolute error.

Tasks and channels. We reimplemented the nine tasks of Heer and Bostock [9], which extend the five original Cleveland–McGill tasks, and added two of our own. The eleven tasks cover seven encoding channels (Tab. 1) and representative stimuli are shown in Fig. 2. Cleveland and McGill place shading and color saturation last in their ranking [4], but neither Heer–Bostock nor the two MLLM replications [6, 13] include a color channel. Task 10 therefore encodes quantity as luminance and Task 11 as saturation, each using two equally sized squares so that no geometric cue is available. Both channels are commonly used in visualizations to encode quantitative values. Because Bendeck and Stasko [2] report color distinctions as a weakness of GPT-4, we expected higher errors on both color tasks than on the geometric tasks.

Stimuli. Following the original protocol, the two compared values are drawn from $s_i = 10 \cdot 10^{(i-1)/12}$ for $i \in \{1, \ldots, 10\}$, a logarithmic series that yields 45 unique value pairs spanning nine ratios from 0.18 to 0.83. Ratios are the controlled quantity because perceptual difficulty depends on the proportion between the compared elements rather than on their individual sizes. Stimuli are 600×600 px PNGs, grayscale on white except for Tasks 10 and 11, where we require color. Following Heer and Bostock [9], the two target elements are labeled A and B, which lets the prompt refer to them unambiguously and keeps responses machine-parsable.

Models. We evaluate six models spanning capability tiers, parameter counts, and architectures: GPT-5.4 and Gemini 3.1 Pro (frontier), Kimi K2.6 (mid), Gemma 4 31B and Gemma 4 26B-A4B (open weights), and Nemotron 3 Nano Omni 30B-A3B (small, open weights). All models were queried through a single unified API with identical stimuli.

Prompt conditions. Every condition is zero-shot with respect to the perceptual judgment: the model is never shown an example of the task, and only the output format is demonstrated. What we vary is how the question is posed by varying three aspects. The first is whether the model is asked for a direct answer or for explicit step-by-step reasoning [15]. The second is whether an expert persona is assigned [16]. The third is whether the instructions appear once or twice in the same prompt [10]. Combining these gives six conditions, listed in full in the supplemental material. All models were

Table 2: Mean LAE per model, lower is better. Models sharing a letter do not differ significantly ($p > 0.05$).

Model	Tier	Mean LAE	
Gemini 3.1 Pro	frontier	0.839	a
GPT-5.4	frontier	2.886	b
Kimi K2.6	mid	3.125	b
Gemma 4 31B	small	3.254	b
Gemma 4 26B	small	3.714	b
Nemotron 3	small	3.958	c

accessed through OpenRouter. Each stimulus was queried once per model and prompt condition at temperature 0.0.

Metric. We report the log absolute error (LAE) used by Cleveland and McGill and by every replication since,

$$\mathrm{LAE} = \log_2\left(|\hat{p} - p| + \frac{1}{8}\right), \tag{1}$$

where $\hat{p}$ is the model's estimate and p the true value, both as percentages. A one-unit decrease in LAE, therefore, corresponds to halving the estimation error, and values are negative whenever the error falls below 0.875 points. The $\frac{1}{8}$ term bounds the metric from below so that an exact answer does not evaluate to $-\infty$.

Reference levels. Because LAE has no natural zero, we compute the score an uninformed responder would assign to our stimuli. Responding uniformly at random over $[0, 100]$ yields an expected LAE of 4.37, and the best possible constant response (31.6%) yields 3.20. We use this to further interpret the measured results.

4 Results

We report three findings about visual encoding channels. Geometric channels produce indistinguishable errors (Sec. 4.2), color channels produce errors worse than guessing (Sec. 4.3), distractor elements degrade estimation while spatial separation does not (Sec. 4.4), and prompt design shifts absolute error without changing the relationship between channels (Sec. 4.5). We first characterize the models themselves (Sec. 4.1) and close with a note on comparing error values across studies (Sec. 4.6). Estimation errors are skewed and heavy-tailed rather than normal ($W = 0.967$, $p < 10^{-6}$), so all tests are non-parametric. We use Kruskal–Wallis with Dunn's post-hoc test for group differences, Mann–Whitney U with Cliff's δ for the prompt axes, and the intraclass correlation (ICC) to measure whether models agree on the same profile of channel errors.

4.1 Model Performance

We first characterize the six models individually (Tab. 2). Five of them result in mean LAE between 2.9 and 4.0. Gemini 3.1 Pro sits far below that range at 0.84, differs significantly from every other model, and is the only model in our set more accurate than the human baseline of 1.25–2.24 [9].

Its advantage coincides with something our prompt conditions do not control. Under the direct-answer prompt, which asks for a quick judgment and nothing more, Gemini produces roughly 4,650 completion tokens per response against 23 for GPT-5.4. It is also the only model whose accuracy remains unchanged when we explicitly request reasoning. However, we cannot separate this from differences in the vision encoder.

Capability tier otherwise predicts accuracy and correlates with API cost almost perfectly ($\rho = 0.94$). Aggregate claims about MLLMs are fragile, since "less accurate than humans" holds for all of our models except Gemini 3.1 Pro. Hence, any claim about channels needs to hold within models, not only across them.

Table 3: Mean LAE per channel, lower is better. Channels sharing a letter do not differ significantly ($p > 0.05$).

Channel	Mean	SD	
Length	2.274	1.949	a
Circular area	2.410	1.811	a
Rectangular area	2.529	1.851	a
Position on a common scale	2.534	2.181	a
Angle/tilt	2.725	1.402	a
Color saturation	5.539	0.916	b
Color luminance	5.602	0.827	b
random responding	*4.373*	*1.500*	

4.2 Geometric Channels are Interchangeable

The five geometric channels are statistically indistinguishable from one another. No pairwise comparison between them reaches significance, while every comparison between a color channel and a geometric channel does at $p < 0.01$ (Tab. 3). The whole geometric range spans 0.45 LAE units. Position on a common scale, the channel humans read most accurately, is no better than angle or area.

This is not a floor effect. All five channels sit well below 4.37, the error expected from random responses, so the models are recovering meaningful quantitative information from each encoding.

Aggregating over models that differ this much could hide per-model structure. However, the six models agree closely on which channels are harder (ICC $= 0.822$), even though they disagree on overall accuracy. Per-model distributions in the supplemental material show overlapping intervals across the geometric tasks for every model. Our significance testing is at the aggregate level, so we make no claim about any individual model.

Human observers do separate these channels, with non-overlapping intervals between position, length, and angle [9], resulting in a total order. Our models do not, and every model except Gemini 3.1 Pro is less accurate than humans.

4.3 Color is not a Quantitative Channel

Both color channels perform worse than guessing. Responding at random would score 4.37, and the best constant answer 3.20, while luminance scores 5.60 and saturation 5.54. Errors this large cannot come from noisy estimation, which would scatter around the truth rather than sit consistently beyond it.

The spread of the responses points in the same direction. Both color channels have a standard deviation near 0.87, against 1.4 to 2.2 for the geometric channels and 1.50 for random responding. The responses are too consistent to be noise and too far from the truth to be estimates. A constant answer of 95 to 97% would reproduce both the mean and the spread we observe, which corresponds to the models reporting that the two squares are roughly equal. Every model is consistent with this, including Gemini 3.1 Pro.

A model that could read pixel values directly would find luminance the easiest channel of all. The vision encoder discretizes the image into patch embeddings before the language model sees anything, so the absolute intensity never reaches it as a number for comparison. We hypothesize that patch embedding does not preserve absolute intensity in a form the language model can compare.

4.4 Surrounding Elements Matter

Two tasks are much harder than others while testing the same channel (Tab. 1). Task 3 splits the bars of a bar chart being compared into two separate groups, whereas Tasks 1 and 2 place them side by side. Task 9 embeds the compared rectangles in a treemap, while Task 8 shows the same two rectangles alone. In both pairs, two things change together. The marks move further apart, and other elements come to lie between them. We call those intervening marks *distractor elements*, since they carry no relevant information.

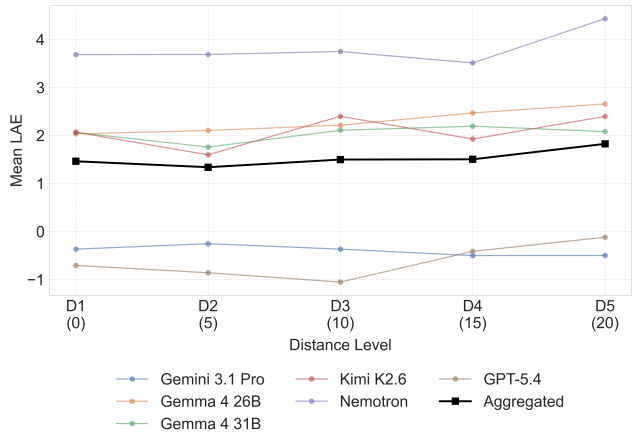


Figure 3: Spatial separation between two compared bars against estimation error, per model and aggregated. Increasing the gap does not systematically degrade proportion estimates.

To separate the two factors, we ran a second experiment with two bars and no distractor elements, varying only the horizontal gap between the two bars. Distance had no significant effect on estimation error ($p = 0.51$, Fig. 3). Separation alone, therefore, cannot explain the Task 3 and Task 9 gaps, which leaves the distractor elements.

Identification is perfect at the two closest spacings and degrades as the gap widens. When the bars are far apart, the model sometimes attaches the labels to the wrong bars and reports the ratio inverted. When it labels them correctly, the ratio is as accurate at the widest spacing as at the narrowest. Distance disrupts what the model reads, not how well it reads it. A design that varies distractor count while holding the marks fixed is the natural next step.

4.5 Prompt Design

Asking for explicit reasoning lowers error, from 3.12 to 2.60 on average, consistent with Guo et al. [6]. The benefit is uneven: GPT-5.4 improves by 1.48 units while Nemotron degrades slightly. Neither an expert persona nor repeated instructions has any measurable effect (Cliff's $\delta = 0.005$ and 0.048). The repetition null matches the scope of the original result, which reports gains for non-reasoning prompts and a neutral effect once reasoning is enabled [10].

What none of these changes is the relationship between channels, which is stable across all six conditions (ICC $= 0.950$). Reasoning prompts make models better at reporting what they perceive. They do not change what the vision encoder makes available to report.

4.6 Comparing Error Values Across Studies

LAE is not scale-invariant. The $\frac{1}{8}$ floor is calibrated for errors in percentage points, as in the original human experiments, so the same underlying accuracy yields values several units apart depending on whether responses are elicited on a $[0,100]$ or a $[0,1]$ scale. Published values for these tasks vary widely. Nguyen et al. [13] report values below the human baseline, where we find values above it, and the response scale is one factor that can account for differences of this size. We therefore explicitly state our own response scale and suggest that comparisons against a cited human baseline confirm that both are expressed in the same units.

5 Implications for Agent-Facing Charts

If a chart is an interface for a model rather than a human, our results suggest five priorities.

No geometric encoding is preferable to another. The advice to prefer position over area rests on differences we do not observe. This widens the design space, but it should not be read as a license: every geometric channel we tested is less accurate than human reading for most models, so quantities that matter should be available as text or data rather than inferred from any encoding.

Never encode a quantity an agent must read in color alone. This is conventional advice for human readers, but here errors on both color channels exceed those of random responding.

Remove elements that do not carry information. Distractor elements degrade estimation while distance does not, so stripping gridlines, decorative marks, and irrelevant series is likely to buy more accuracy than any choice among geometric encodings. Separation is safe for estimation but not for identification, so where marks must be labeled, keeping the label and mark close reduces the risk that the model reads the wrong one.

Model choice matters more than encoding choice. The gap between Gemini 3.1 Pro and the next model (2.05 LAE units) is more than four times the spread across geometric channels (0.45). Accuracy tracks API cost ($\rho = 0.94$). The channel profile is shared across models (ICC $= 0.822$), so a stronger model does not make color readable.

Prompting helps estimation unevenly and cannot rescue a channel. Requesting reasoning lowers mean error by 0.52 units, ranging from a 1.48-unit gain (GPT-5.4) to a slight loss (Nemotron). Personas and repeated instructions had no measurable effect. The channel behavior is stable across prompts (ICC $= 0.950$).

6 Limitations

Six models are a small sample for claims about capability tier and architecture, and each condition was run once, so run-to-run variance is unmeasured. Our stimuli are deliberately minimal and grayscale, which differs from the colored-segment stimuli used by Guo et al. [6]; since they report that removing fill color substantially lowers accuracy, our absolute errors should be read as a conservative bound rather than a typical operating point. Interchangeability is a null result: we did not detect differences between the geometric channels, which is not the same as establishing that none exist, though the consistency across six models and six prompt conditions bounds how large any real difference could be. The instruction-repetition condition was applied only to the two smallest models, so its effect cannot be separated from the identity of those models, and we draw no conclusion from it beyond the two models tested.

7 Conclusion

Across six multimodal LLMs, the geometric encoding channels that human graphical perception carefully separates are interchangeable, and the color channels that humans read poorly are not read as quantities at all. Only Gemini 3.1 pro achieves better-than-human accuracy. Within the geometric encodings, what predicts error is not the channel but the extent of competing visual material surrounding the marks being compared. For visualizations destined to be read by agents, that reorders the design checklist: remove clutter first, avoid color encodings second, and treat the choice among geometric channels as largely immaterial. Our stimulus generator and evaluation harness are seeded and released, so the same measurements can be repeated on models released after this study.

Supplemental Materials

They include (1) the stimulus generation pipeline and evaluation, seeded for exact reproduction of all images; (2) the complete stimulus set with per-task layout constraints and grouping tables; (3) the six full prompt texts; (4) raw model responses and parsed results; (5) per-model channel and task profiles; (6) the chance-level baseline computation; and (7) the recomputation script for the statistical comparison.

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